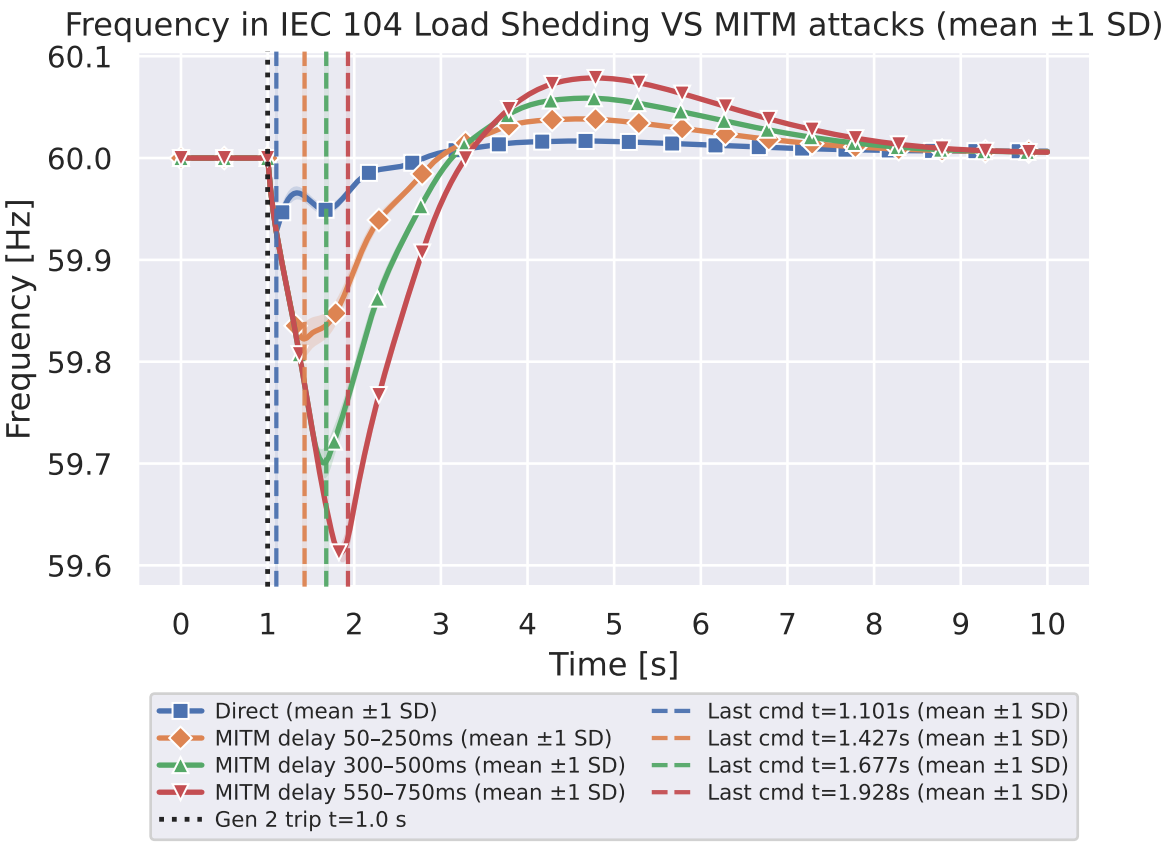
Frequency in IEC 104 Load Shedding VS MITM attacks (mean ±1 SD)
Frequency [Hz]
60.1
60.0
59.9
59.8
59.7
59.6
0
1
2
3
4
5
6
7
8
9
10
Time [s]
Direct (mean ±1 SD)
MITM delay 50–250ms (mean ±1 SD)
MITM delay 300–500ms (mean ±1 SD)
MITM delay 550–750ms (mean ±1 SD)
Gen 2 trip t=1.0 s
Last cmd t=1.101s (mean ±1 SD)
Last cmd t=1.427s (mean ±1 SD)
Last cmd t=1.677s (mean ±1 SD)
Last cmd t=1.928s (mean ±1 SD)